

Neuropathology and Functional Deficits in a Model of Birth Asphyxia in the Precocial Spiny Mouse (*Acomys cahirinus*)

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Key Words

Brain damage · Brain hypoxia · Brain injury · Neonatal brain injury · Birth asphyxia · Behavioural outcomes

Abstract

Birth asphyxia can result in sensory impairment, learning and memory deficits without gross brain injury and severe motor deficits. We developed a model of birth asphyxia resulting in mild neurological injury and cognitive impairment using a long-gestation species with precocial fetal development. Spiny mice (*Acomys cahirinus*) underwent caesarean-section delivery or 7.5 min of asphyxia at 37 days gestational age (term is 39 days). Brain histology was examined at 1 and 7 days of age, and behaviour was evaluated to 28 days of age. Asphyxiated offspring showed significant impairment in non-spatial memory and learning tasks, accompanied by central nervous system inflammation and increased apoptotic cell death but without the presence of large necrotic or cystic lesions.

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Introduction

Birth asphyxia has been associated with an increased risk of cerebral palsy, and also with impaired cognitive function in school-age children, in the absence of gross motor deficits. The incidence of these impairments is difficult to determine as studies of long-term outcomes of neonatal encephalopathy, or of the risk factors associated with perinatal hypoxia-ischaemia, use different inclusion criteria to implicate the involvement of hypoxic and asphyxic events in the perinatal period [1, 2]. Studies show that birth asphyxia may have been involved in the development of functional impairments in up to 50% of children with neonatal encephalopathy [3]. A recent meta-analysis has shown that the extent of encephalopathy correlates with the degree of functional impairment, with severe encephalopathy resulting in impairment in every cognitive domain whilst more moderate brain damage resulted in more heterogeneous functional outcomes [1]. Cognitive impairments associated with birth asphyxia include impairments of speech, hearing, vision, memory and learning difficulties, altered behaviour and emotional problems including hyperactivity and anxiety problems [3–7]. Term birth asphyxia resulting in moderate encephalopathy in the absence of any gross motor impairment resulted in lower IQ, and reading and spelling skills in 8-year-old school children [7].

Hypoxia-ischaemia in the perinatal period is known to result in central nervous system (CNS) inflammation (characterized by the activation of microglia and infiltration of macrophages) [8], cell death via necrotic and apoptotic mechanisms [9] and astrogliosis (an increase in astrocyte number accompanied by an increase in glial fibrillary acidic protein (GFAP) expression) [10, 11]. Whilst it is difficult to assess the magnitude of these histological components of brain injury in humans, as their direct imaging is difficult, it is proposed that these cellular responses occur in infants subjected to perinatal asphyxia, even in the absence of gross lesions or the appearance of major musculoskeletal disability.

It is difficult to determine the most appropriate animal model to be used for the assessment of perinatal brain injury. Primates are expensive, and ethical considerations constrain their use as a research tool. Altricial species (such as rats and mice) require manipulations to be performed *ex utero*, when some aspects of brain maturation, particularly the onset of myelination, have reached comparable development to that of the human infant at birth. As a result, a number of these rodent studies use models of focal ischaemia [12] which do not replicate the global hypoxic-ischaemic events that occur in the clinical situation. Thus, in order to investigate an *in utero* condition using an *in utero* model of fetal hypoxia, we have used the precocial spiny mouse. It has been suggested elsewhere that this species is particularly suited to the assessment of brain pathways subserving the postnatal development of behaviour [13]. Unlike other rodent species, but like the human fetus, brain growth in the spiny mouse is greatest at around the time of birth [unpubl. obs.], and myelination begins in the latter part of gestation and continues into the early postnatal period.

The aim of this study was to develop a model of near-term birth asphyxia in a precocial species that results in a degree of brain damage compatible with survival of the neonate, and in which functional alterations in the absence of severe motor deficits can be assessed.

Methods

Animals and Experimental Protocol

All experiments were approved by the Monash University, School of Biomedical Science Animal Ethics Committee and were conducted in accordance with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes.

The spiny mice used in this study were obtained from our own laboratory colony, time-mated and housed as previously described [14]. On day 37 of gestation, pregnant dams were assigned to either delivery by caesarean section (C-section), or a birth as-

phyxia delivery using a protocol previously developed by us [15]. Briefly, pregnant dams were killed by cervical dislocation on day 37 of gestation, and the pups were either delivered immediately from the uterus (C-section control delivery, $n = 16$ dams, 53 offspring) and resuscitated, or the pregnant uterus isolated and placed in a saline bath at 37.5–38°C, and the pups delivered after 7.5 min and then resuscitated (birth asphyxia group, $n = 19$ dams, 59 offspring). This duration of *in utero* asphyxia was chosen because Ireland et al. [15] found that 5 min had no detectable effects whereas 10 min resulted in 100% death of the pups. For each group, when the uterus was opened, the pups were removed within 30 s, their mouths cleared of membranes or mucus, and they were stimulated to breathe by intermittent gentle palpation of the chest and abdomen for 5–10 min using a moist cotton tip. They were allowed to recover for 1–1.5 h in warm bedding taken from the cross-foster cage, and then introduced into the cage of the cross-foster dam which had given birth within the preceding 24 h; the cross-foster dam's own pups were removed at this time. The control group consisted of pups that were expelled from the uterus immediately after it had been removed from the abdomen, and were then resuscitated and cross-fostered in the same manner as the birth asphyxia pups.

The offspring were randomly assigned to undergo post-mortem at either 1, 7 or 30 days of age. Litters of C-section and asphyxiated pups which were assigned to undergo post-mortem at 1 day (C-section $n = 2$ dams and a total of 7 offspring; asphyxia $n = 6$ dams and a total of 18 offspring), or 7 days (C-section $n = 5$ dams and a total of 18 offspring; asphyxia $n = 3$ dams and a total of 9 offspring), and they were left undisturbed in the cross-foster cage until post-mortem. Behavioural analysis was done with the animals surviving until 30 days of age. The relationship of outcomes to gender was not formally analysed in this study because of the relatively small number of animals studied at 30 days of age (C-section: 4 males, 5 females; asphyxia: 4 males, 6 females).

At the scheduled time of post-mortem examination, surviving pups were killed by decapitation and the brains collected. Each brain was fixed in 10% phosphate-buffered formalin for 36 h, sectioned along the mid-line into right and left hemispheres, paraffin embedded, and cut into 5- μ m-thick sections in the sagittal plane for immunohistochemistry and histological analyses as outlined below.

Immunohistochemistry and Histological Analyses

For all immunohistochemical analyses, sections were de-waxed in xylene and rehydrated in graded ethanol. All incubations were performed at room temperature, and phosphate-buffered saline (PBS; 0.1 M pH 7.4) was used for all washes unless otherwise stated. All sections were dehydrated in graded ethanols and xylene prior to applying DPX mounting medium and a cover slip.

Microglia and macrophages were stained using lectin histochemistry [16]. Antigen recovery was performed by heating the sections in 0.01 M citric acid buffer (pH 6) for 3 $\times$ 5 min using a microwave; sections were then allowed to cool in the hot buffer for 20 min prior to washing. Endogenous peroxidases were blocked by incubating the sections in 0.3% hydrogen peroxide in 50% methanol in distilled water for 15 min. Sections were washed, pre-incubated in 5% normal goat serum (NGS), 2% BSA in PBS with 0.1% Triton X-100 for 20 min, and then incubated in peroxidase-labelled isolectin B4 derived from *Bandeiraea simplicifolia*

(lectin; Sigma Aldrich, USA, 1:200 dilution) overnight at 4°C. Following thorough washing, staining was visualized using diaminobenzidine (DAB, Pierce, USA, 1:10 dilution).

Astrocytes were stained using GFAP immunohistochemistry. Sections underwent antigen retrieval by heating in citric acid buffer for 3 × 3 min using a microwave oven and were then left to cool in the buffer for a further 15 min. They were then washed and incubated in 3% hydrogen peroxide in 50% methanol in distilled water for 20 min, prior to incubation in a blocking solution comprised of 5% normal rabbit serum in PBS for 30 min. Sections were then incubated overnight at 4°C in mouse anti-GFAP primary antibody (Sigma Aldrich, USA, 1:400) diluted in PBS with 2.5% NGS and 0.3% Triton X-100. The following day, sections were washed and incubated in biotinylated rabbit anti-mouse secondary antibody for 1 h. Sections were again washed and incubated in streptavidin horseradish peroxidase (1:200 diluted in PBS) for 1 h. Staining was visualized with DAB (1:10).

Cells in which the apoptotic cell death cascade had been induced were identified using immunohistochemical staining for activated caspase-3 [17]. Sections were first heated in citric acid buffer for 3 × 5 min using a microwave oven, allowed to cool in the buffer for a further 20 min, and then incubated in 0.3% hydrogen peroxide diluted in 50% methanol and 50% distilled water for 20 min. Sections were again washed prior to a 1-hour incubation in 5% NGS, 2% BSA in 0.1% Triton X-100 in PBS, and then incubated overnight at 4°C in rabbit anti-activated caspase-3 primary antibody (R&D Systems, USA, 1:1,000 diluted in PBS with 5% NGS and 2% BSA). The specificity of this antibody in spiny mouse tissue has been shown by us previously using Western blotting techniques. The following day, sections were washed and incubated in biotinylated goat anti-rabbit secondary antibody for 1 h, washed and incubated for 1 h in streptavidin horseradish peroxidase (1:200 diluted in PBS), and staining visualized using DAB (1:10).

Quantitation of Immunohistochemistry Results

The CA1 and CA3 regions of the hippocampus, inferior colliculus, corpus callosum, cerebellar white matter, granular layer, molecular layer and medial nucleus were examined at 1 and 7 days of age. All immunopositive cells were counted in 3 fields of view per brain region on 2 brain sections per animal, and the average count from these 6 fields of view then calculated for each animal. Activated caspase-3 staining was expressed as the percentage change from C-section-delivered age-matched offspring. The histological data were obtained in a blinded manner as animals were only identified by numbers and were not distinguishable by treatment group until the statistical analysis was done.

Behavioural Testing

All animals assigned to undergo post-mortem at 30 days of age underwent behavioural testing between days 1 and 28 of age. The data exclude any from those animals who did not reach 30 days, which includes pups that died after attack by the cross-foster parents during this time.

Open-Field Testing

At approximately 24 h of age (day 1; day of delivery is day 0), and every 3 days thereafter until 28 days of age, pups were placed in an open field to assess spontaneous locomotor activity and other behaviours. Pups in their home cage (containing a cross-foster

dam, an aged-matched adult male and littermates) were habituated to the room in which the open-field apparatus was kept for at least 2 h prior to testing. Open-field testing was conducted between 11.00 and 13.00 h. On days 1 and 4, the open-field size was 24 × 24 × 22 cm (L × W × H) and the trial duration was 10 min. Each trial after these days was for 15 min in an open field 50 × 50 × 50 cm (L × W × H). Field size and trial length were decided upon after monitoring the behaviour of vaginally delivered spiny mouse pups at comparable ages, which showed that younger pups travelled a lesser distance when placed in a large, rather than a small open field. Lighting levels were 2.8 lx for all trials.

Limelight (Neuroscience Inc., Japan) acquisition software was used to track the movement of the pups throughout the open-field trial. Images were captured at a rate of 7.5 Hz and stored on disk. Post-acquisition analysis allowed movement of the animal's body throughout the testing period to be followed, so that the total distance travelled (cm), and time spent (seconds) in the central zone (where the 8-cm outer perimeter is excluded from the area of analysis) versus outer zone, were determined by the dedicated software. A number of parameters were manually scored by an observer using the software's behavioural analysis scoring system, including: (1) percentage of time spent immobile, defined the percentage of the total trial in which movement of the animal's body of less than 0.5 cm for 5 consecutive frames; (2) velocity of movement (cm/s), calculated from the total distance travelled during the time (s) spent mobile. Time spent mobile was determined by the period of time within a trial in which the animal was not mobile. The number of bouts of the following behaviours was manually scored by an investigator: (3) grooming, defined as an immobile animal using one or two forepaws to contact the body, face or whiskers; (4) wall jumps, when only the hind limbs remain on the ground, with the forelimbs raised on the wall, and (5) wall flips, when an animal has no limbs in contact with the floor of the open field. This latter behaviour is characteristic of the spiny mouse after approximately 10 days of age in both novel and familiar environments (e.g. the home cage).

Novel Object Recognition Test

The Novel Object Recognition Test (NORT) assesses non-spatial memory deficits by exploiting the innate exploratory behaviour of rodents [18]. On day 28, the 15-min open-field trial outlined above also serves as a habituation period for the NORT to reduce the influence of anxiety and stress on the outcome. Immediately following the completion of the open-field test, the animal was returned to its home cage for 5 min whilst 2 plastic children's blocks of identical shape, colour and size were placed in the open field. Blocks were positioned 6 cm from the walls of the cage to ensure an unobscured view of the animal at all times. The animal's behaviour was recorded for a 30-min period, which serves as a learning period in which the animal acquires information regarding the objects in the open field. Following this period, the animal was returned to its home cage for a further 1 h (retention period). During this time, one of the two plastic blocks was replaced with a plastic block of a different shape and colour (the novel object). The remaining plastic block (which had been explored previously in the learning trial) and the novel object were wiped down with 70% ethanol to remove olfactory cues. The animal's behaviour and exploration of the two objects were recorded for the next 30 min. In the final NORT trial (termed the 'recall

Table 1. Effect of birth asphyxia on survival of spiny mouse pups, body weight and brain weight at 1, 7 and 30 days after delivery

	1 day		7 days		30 days		Two-way ANOVA p value
	C-section	asphyxia	C-section	asphyxia	C-section	asphyxia	
Survival at time of PM, % (viable/cross-fostered, n)	100 (7/7)	53.33 (8/15)	58.82 (10/17)	88.88 (8/9)	32.14 (9/30)	37.01 (10/17)	
Body weight, g (1 h after delivery)	4.804 ± 0.073	4.945 ± 0.138	5.077 ± 0.086	4.949 ± 0.099	5.239 ± 0.112	4.857 ± 0.125	P _{birth} = NS P _{age at PM} = NS P _{interaction} = NS
Body weight, g (at PM)	4.056 ± 0.100	4.504 ± 0.272	7.447 ± 0.578	8.267 ± 0.228	23.398 ± 0.779	21.776 ± 0.490	P _{birth} = NS P _{age at PM} ≤ 0.001* P _{interaction} = 0.0356
Brain weight, g	0.435 ± 0.013	0.434 ± 0.033	0.575 ± 0.030	0.538 ± 0.019	0.739 ± 0.008	0.707 ± 0.014	P _{birth} = NS P _{age at PM} ≤ 0.001* P _{interaction} = NS

* Significant difference between C-section and asphyxiated pups. Two-way ANOVA with Bonferroni post-hoc tests.

trial'), the nose, rather than the body of the animal was tracked using Limelight software. When the animal's nose was pointed at, and within 2 cm of the block, this was deemed to be active exploration of the block. The total time spent exploring each block in the first 10 min of the trial was used to calculate the discrimination index. The discrimination index is calculated by subtracting the time spent exploring the familiar object from the time spent exploring the novel object, divided by the total time spent exploring both objects [18]. Thus, a discrimination index of 1 indicates that the animal spent all of the time exploring the novel object, 0 indicates no preference for either object, and -1 represents all time spent exploring the familiar object.

Rotarod

On day 10, animals underwent rotarod testing 30 min after the open-field test. All animals were allowed 2 learning/habituation trials on the rotarod. After the learning trials were completed, the latency to fall was recorded for 5 trials (T1–T5). The initial two tests (T1 and T2) were performed immediately after the last learning trial. Animals were then allowed an 8- to 10-min rest period after the second trial and then underwent a further three trials (T3–T5).

Statistical Analyses

All data are expressed as means ± SEM. All data met assumptions of normality and homogeneity of variances. Lectin and GFAP cell counts were analysed using two-way ANOVA (method of birth × age at post-mortem) followed by one-way ANOVA (treatment) per time of tissue collection. The relative (percent) change in expression of activated caspase-3 protein expression in the C-section-delivered pups was compared to asphyxiated pups at 1 or 7 days of age, by independent t tests. For the behavioural data, total distance travelled, percentage of total distances travelled in the central versus outer zones of the field, number of bouts of wall jumps, wall flips, and grooming were analysed using two-way repeated-measures ANOVA (mode of birth × age at testing). Comparisons were made for data at 1 and 4 days of age, and at 13,

22 and 28 days; the difference in trial time and field size at days 1 and 4, and at the later ages, obviated comparisons between all ages. For the rotarod data, mean, median, minimum and maximum latencies to fall were calculated for each animal over all 5 trials, and then for the 3 and 4 best trials to remove effects of 1 or 2 poor trials. Results between the treatment groups (C-section vs. birth asphyxia) were then compared using independent t tests, and also using 2-way repeated-measures ANOVA (method of birth × trial number) to investigate effects of treatment and trial on latency to fall. The averages of the 2 pre-rest trials (T1 and T2) were compared to the averages of the 3 post-rest trials (T3–T5) between treatment groups. Post-hoc Bonferroni comparisons were then used to determine differences between treatment groups. For the NORT data, discrimination indexes for C-section-delivered and asphyxiated offspring were compared using the independent-samples t test.

Results

Offspring Survival, Organ and Body Weights

There were no differences in litter size, maternal weight, and placental weights for animals assigned to the C-section or asphyxial delivery groups (data not shown). Of the C-section-delivered pups, 98.1% (52/53) were alive 1 h after delivery, and were then placed in a cross-foster cage (the pup that died was from a litter of 5, where at least 1 neonatal death is not unusual). After the asphyxial delivery, 86.4% (51/59) of pups were alive at 1 h and were presented to a cross-foster dam. At this postnatal age (1 h), and at 1, 7 or 30 days of age, there were no differences in weights of the pups delivered by C-section or asphyxia (table 1). At the conclusion of behavioural test-

ing at 30 days of age, post-mortem examination showed that there were no significant differences in weights of the major organs other than the brain in the C-section and asphyxiated groups.

Despite attempts to minimize rejection by exposing the offspring to soiled bedding prior to introducing pups to the cross-foster dam in her cage, rejection rates were high, but not significantly different between the C-section and asphyxiated groups (table 1); Kaplan-Meier survival curves for the two groups of pups are shown in figure 1. Pups were either found dead in the cage, sometimes with obvious signs of attack, or were not found at all, and presumably had been cannibalized by the dam or the companion male. The majority of dead pups were found within the first 48 h after delivery.

Neuropathology at 1 and 7 Days after Delivery CNS Inflammation

Birth asphyxia increased the number of lectin-positive cells in the CA1 [$F(1, 27) = 18.81, p < 0.001$] and CA3 [$F(1, 27) = 16.73, p < 0.001$] regions of the hippocampus, inferior colliculus [$F(1, 27) = 6.874, p < 0.050$], corpus callosum [$F(1, 27) = 9.136, p < 0.010$] (fig. 2). Lectin-positive cells with microglia-like morphology (fig. 2e) were seen in all brain regions examined. A smaller number of cells showed macrophage-like morphology (fig. 2f), indicated by the absence of processes and uniform size and shape. These were noted to be present particularly in the corpus callosum in most asphyxiated animals at 7 days of age (fig. 2d), but were rarely seen in brains from C-section offspring. In the hippocampal CA1 and CA3 regions, the number of lectin-positive cells was increased at both 1 and 7 days in the birth asphyxia pups, compared to the C-section pups. In the cerebellum, asphyxia increased the number of lectin-positive cells in the granular layer [$F(1, 25) = 4.653, p < 0.050$; fig. 2], but there was no effect of the mode of delivery in the molecular layer [$F(1, 25) = 7.140, p > 0.05$] and medial nucleus [$F(1, 25) = 6.965, p > 0.05$]. At 1 week of age (but not at day 1), there was a significant increase in the number of lectin-positive cells in the inferior colliculus, corpus callosum and granular layer of the cerebellum. The number of lectin-positive cells in the molecular layer and medial nucleus of the cerebellum increased between 1 and 7 days of age.

Glial Fibrillary Acidic Protein

Birth asphyxia increased the number of astrocytes in the CA1 region of the hippocampus [$F(1, 27) = 4.817, p < 0.050$], corpus callosum [$F(1, 27) = 13.07, p < 0.010$] and

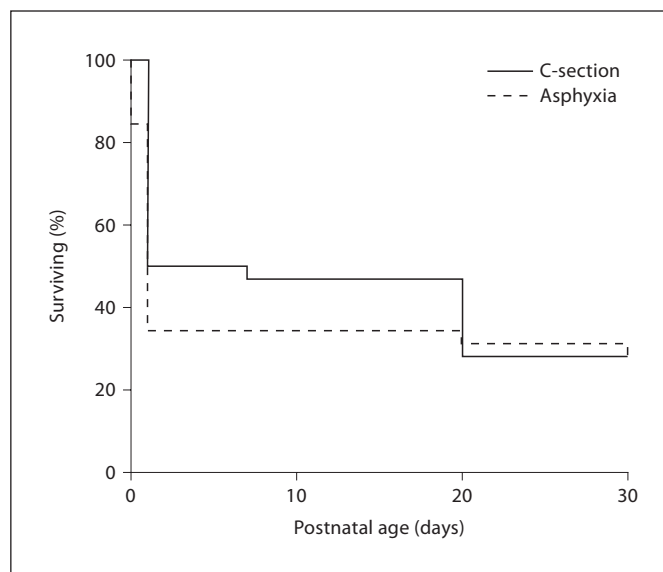


Fig. 1. Kaplan-Meier survival curves for C-section-delivered (solid line) and asphyxiated (dotted line) pups to 30 days of age.

granular layer of the cerebellum [$F(1, 25) = 7.833, p < 0.010$] (fig. 3). In the corpus callosum, the increased number of astrocytes was evident at 1 and 7 days of age following birth asphyxia, whereas an increase in the hippocampal CA1 was not evident until 7 days of age. In the granular layer of the cerebellum, a significant increase in the number of GFAP-positive cells at 1 day of age had resolved to levels not different from C-section-delivered offspring at 7 days of age. The number of astrocytes in the inferior colliculus, hippocampal CA3, and cerebellar medial nucleus increased with age, but there was no effect of birth asphyxia in these regions.

Apoptosis

Reasonably large numbers of cells immunopositive for activated caspase-3 were present throughout the brains of the C-section-delivered pups at both 1 and 7 days of age (fig. 4). The morphology and size of these cells suggest that both neurons and astrocytes were undergoing a physiological, or constitutive, apoptosis during postnatal brain development in this species. Birth asphyxia resulted in an increase in the number of cells in the corpus callosum expressing this apoptotic protein on days 1 and 7, and in the molecular layer of the cerebellum on day 7 (fig. 4).

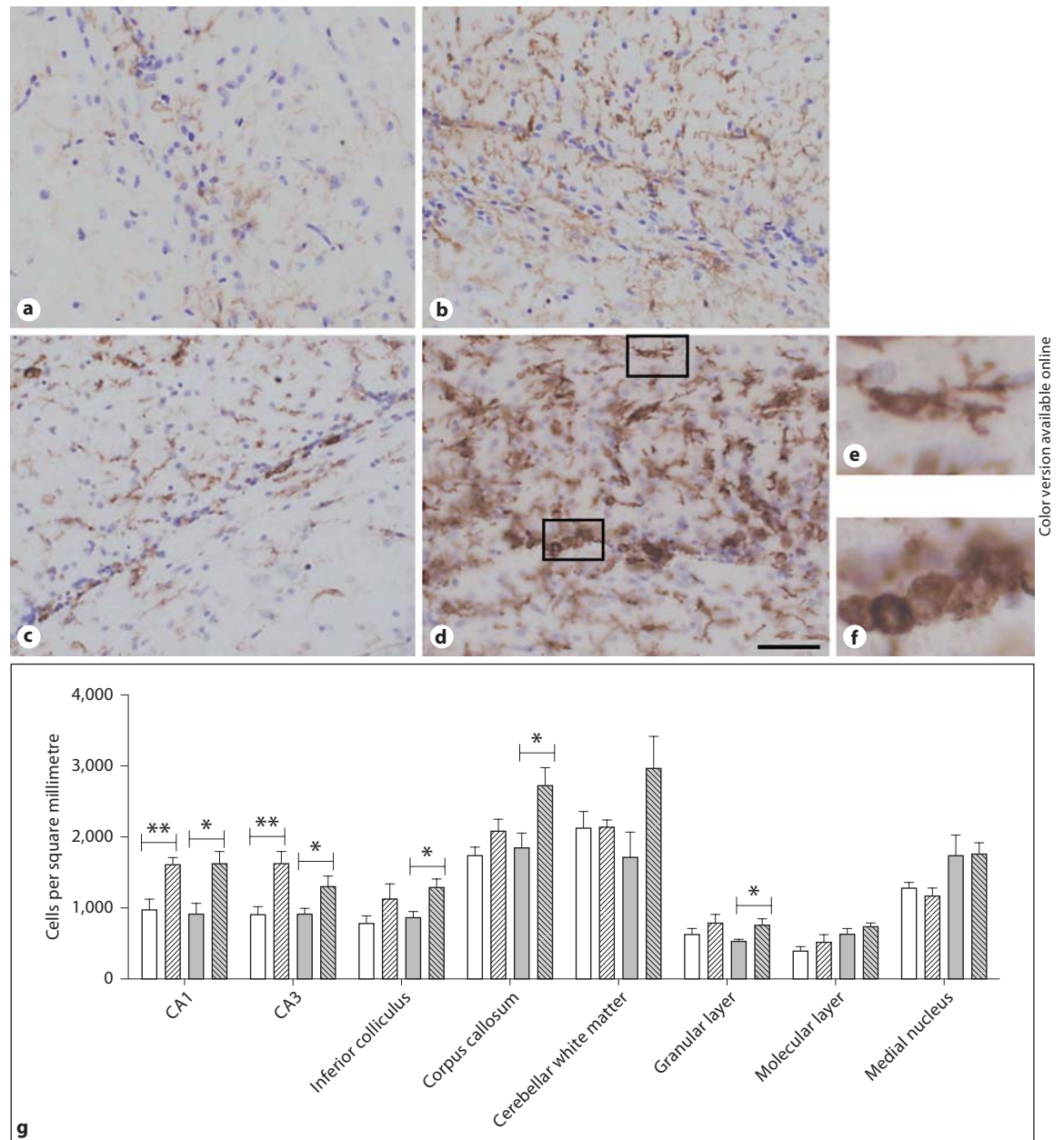


Fig. 2. Lectin staining in the corpus callosum from C-section-delivered pups (**a, c**) and asphyxiated pups (**b, d**) at 1 (**a, b**) and 7 (**c, d**) days following birth asphyxia. Note the typical morphology of microglia (**e**; magnification of framed area in **d**) and macrophages (**f**; magnification of framed area in **d**) in the corpus callosum of an animal showing a significant inflammatory response 7 days

following asphyxiated delivery (**d**). Scale bar = 50 μ m (**a–d**). Number of lectin-positive cells per square millimetre at 1 (white bars) and 7 days (grey bars) following C-section delivery (open bars) or asphyxiated delivery (hashed bars) (**g**). Two-way ANOVA with one-way ANOVA (method of delivery) per time at tissue collection. * $p \leq 0.050$, ** $p \leq 0.010$.

Functional Outcomes of Birth Asphyxia Open Field

The total distance travelled in the open field was similar for both the birth asphyxia and C-section-delivered offspring at 1, 4, 13, 22 and 28 days of age (data not shown).

There was no significant effect of birth asphyxia on thigmotaxis (wall-hugging behaviour) at any of these ages (fig. 5c, e, f), with no differences in the time spent in the central zone of the open field, or the distance travelled in the central zone of the open field at any age (data not

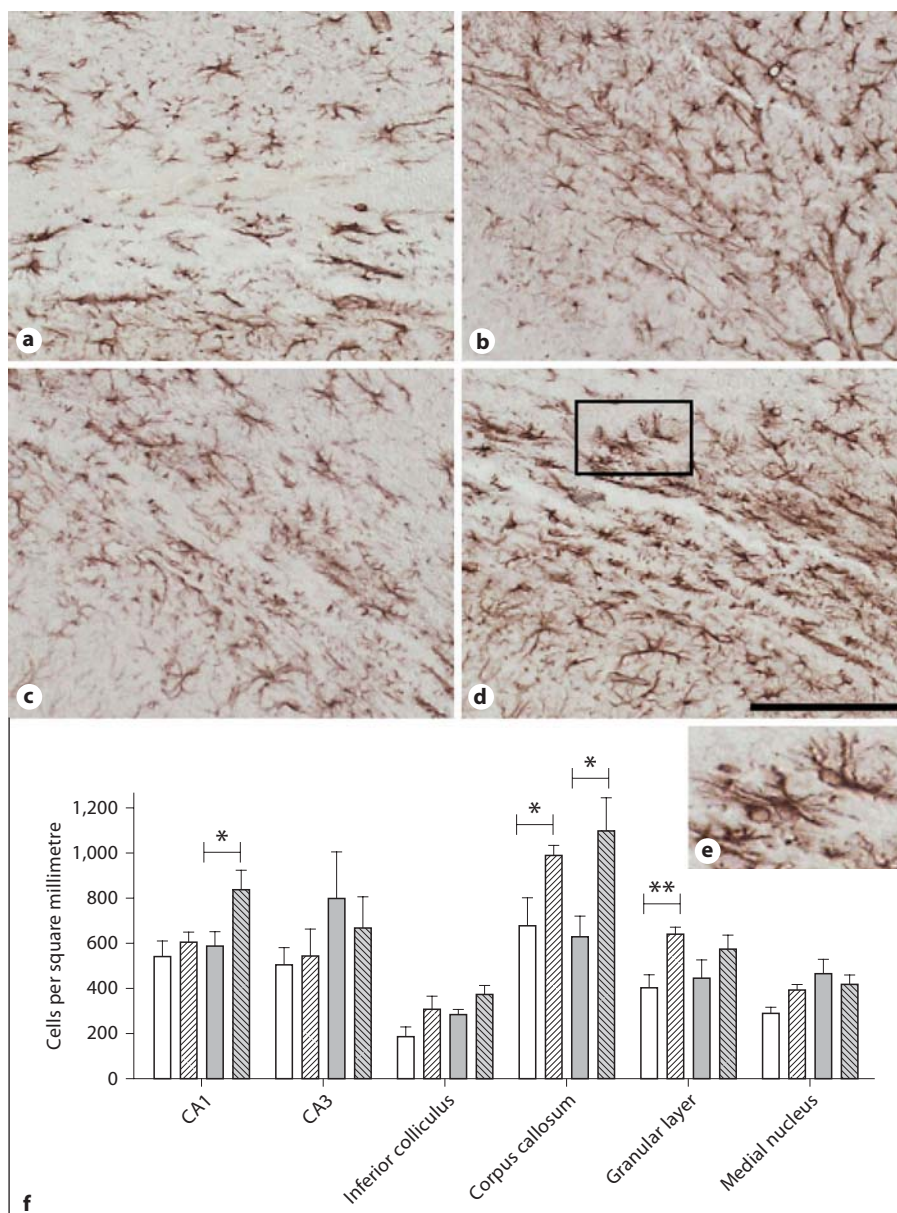


Fig. 3. GFAP staining in the corpus callosum from C-section-delivered pups (**a, c**) and asphyxiated pups (**b, d**) at 1 (**a, b**) and 7 (**c, d**) days following birth asphyxia. Note the typical morphology of activated astrocytes (**e**; magnification of framed area in **d**). Scale bar = 100 μ m (**a–d**). Number of GFAP-positive cells per square millimetre at 1 (white bars) and 7 days (grey bars) following C-section delivery (open bars) or asphyxiated delivery (hashed bars) (**f**). Two-way ANOVA with one-way ANOVA (method of delivery) per time at tissue collection. * $p \leq 0.050$, ** $p \leq 0.010$.

shown). There was no significant difference in the times spent mobile or stationary, or in the velocity of travel when actually moving (data not shown). However, the birth asphyxia pups showed a significant reduction in the number of bouts of grooming on day 4 and a near-significant reduction in the number of wall jumps on day 28 ($p = 0.060$) (fig. 5). There was no significant difference in the number of wall flips (all limbs off the ground) at 28 days (fig. 4c), a behaviour that is not seen at 1 and 4 days of age.

Rotarod

At 10 days of age, the time spent on the rotarod, whether calculated as the mean or median time over all 5 trials, was significantly shorter in asphyxiated offspring than in C-section-delivered offspring (fig. 6a, b). The shortest and longest trial recorded for each animal was also significantly lower in asphyxiated offspring (fig. 6c, d). The mean of the 3 and 4 longest trials for each animal was also significantly reduced in asphyxiated offspring compared with C-section-delivered controls (fig. 6d, e). Repeated-measures ANOVA (method of delivery $\times$ trial number)

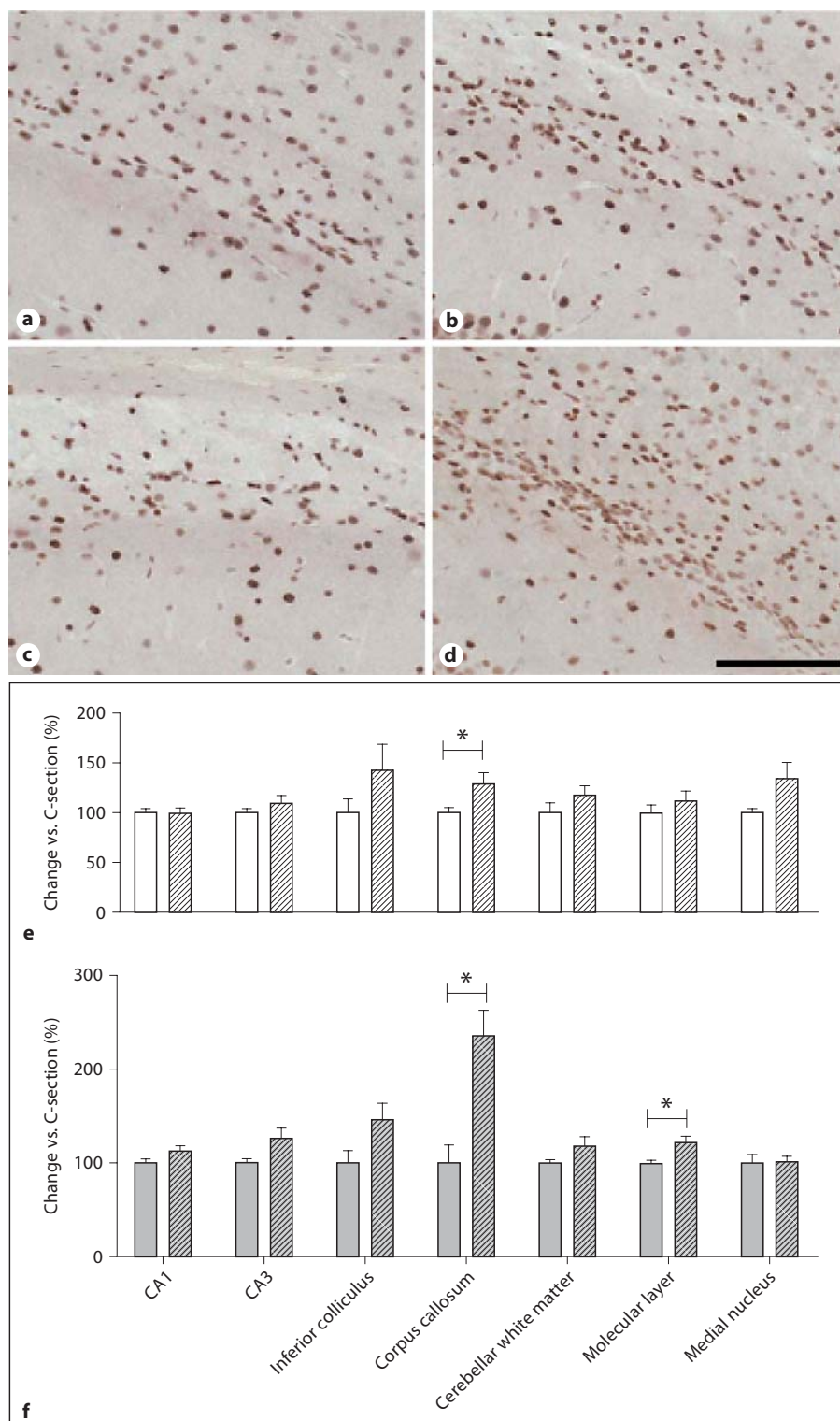


Fig. 4. Activated caspase-3 staining in the corpus callosum from C-section-delivered pups (**a, c**) and asphyxiated pups (**b, d**) at 1 (**a, b**) and 7 (**c, d**) days following birth asphyxia. Scale bar = 100 μ m (**a-d**). Percentage change in number of activated caspase-3-positive cells from C-section-delivered pups (within time point) at 1 (white bars, **e**) and 7 days (grey bars, **f**) following C-section delivery (open bars) or asphyxiated delivery (hashed bars). Independent-measures t test. * $p \leq 0.050$.

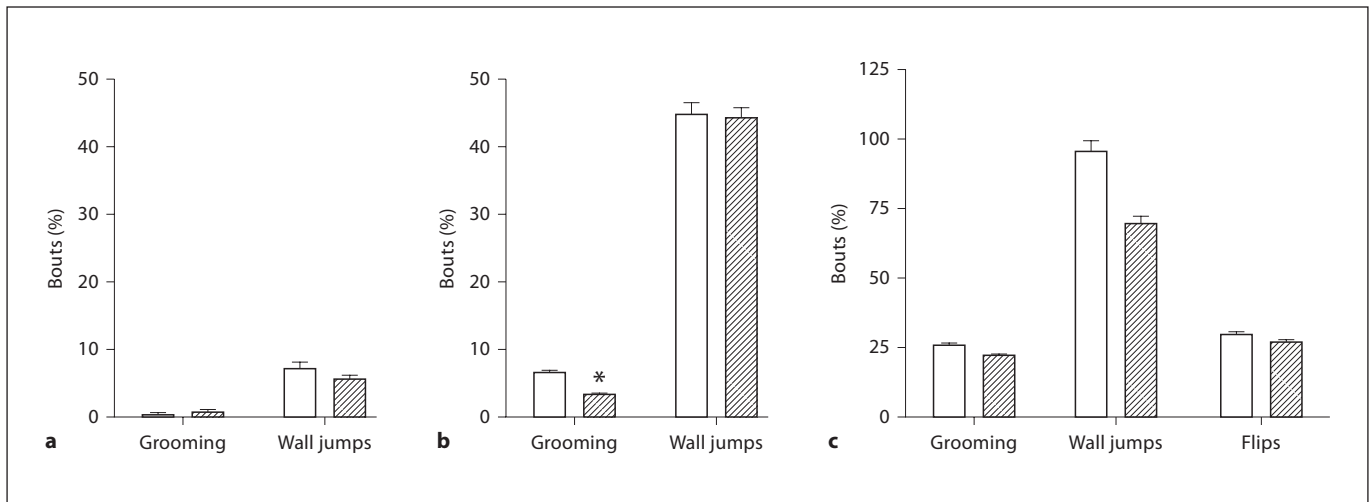
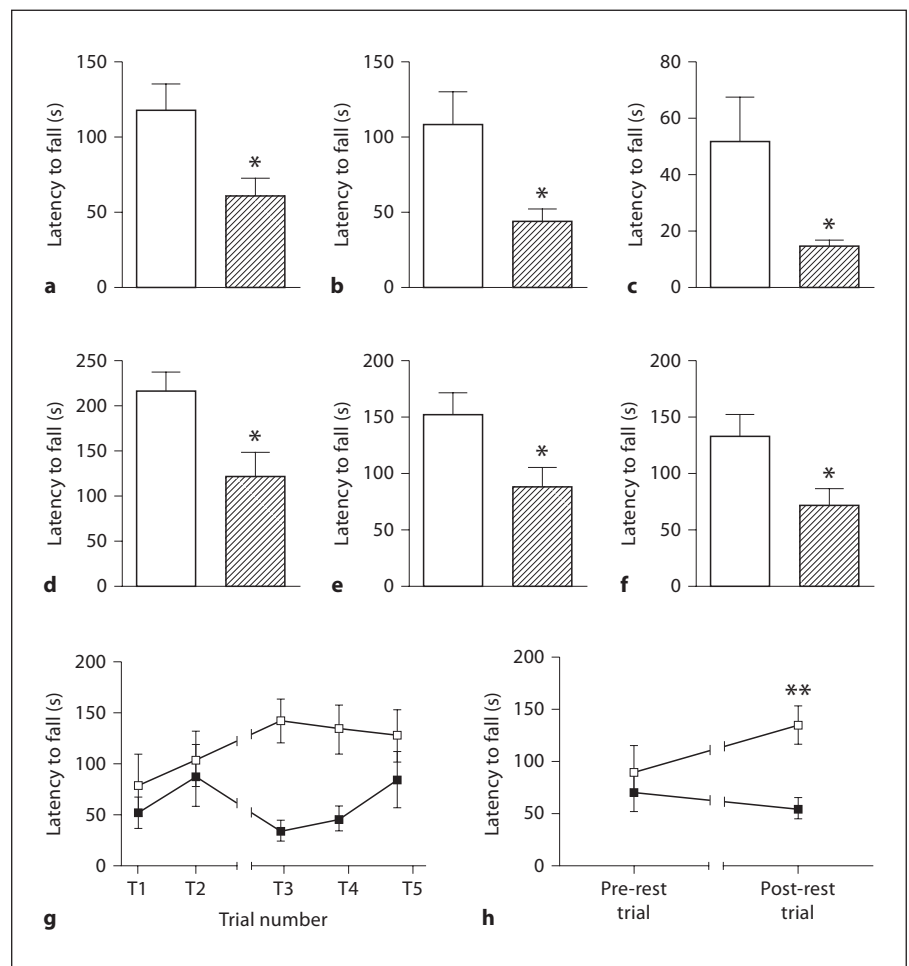


Fig. 5. Number of bouts of grooming and wall jumps at 1 (**a**), 4 (**b**) and 28 (**c**) days of age. Wall flips were not seen at 1 and 4 days of age. Bouts of wall jumps at 28 days of age (**c**). Two-way repeated-measures ANOVA (age at testing $\times$ method of delivery) with Bonferroni post-hoc tests. * $p \leq 0.050$.

Fig. 6. Mean (**a**), median (**b**), minimum (**c**), maximum (**d**), mean of 3 best trials (**e**), mean of 4 best trials (**f**) over all rotarod trials. There was a significant reduction in the latency to fall in each of these parameters. Latency to fall by trial number (**g**) demonstrated a significant effect of method of delivery. Note the break in the x-axis which represents the 10-min rest period between T2 and T3. Mean of the pre-rest trials (T1 and T2) and mean of the post-rest trials (T3–T5) shows a significant reduction in latency to fall in asphyxiated animals only after the rest period (**h**). Independent-measures t test (**a–f**). Two-way repeated measures ANOVA (trial number $\times$ method of delivery) (**g**). Two-way repeated-measures ANOVA (pre-/post-rest $\times$ method of delivery) with Bonferroni post-hoc tests (**h**). * $p \leq 0.050$, ** $p \leq 0.010$.



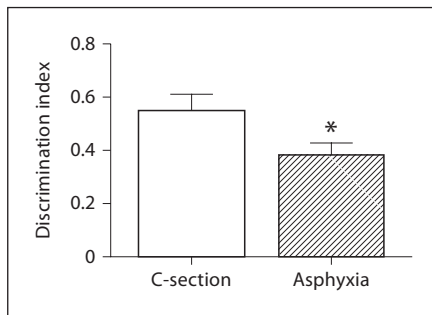


Fig. 7. Discrimination index in the NORT for C-section-delivered and asphyxiated offspring at 28 days of age. Note the significant reduction in the discrimination index of asphyxiated offspring. Independent-measures *t* test. * $p \leq 0.050$.

showed that there was a significant overall effect of the method of delivery ($p = 0.017$), but no effect of trial number ($p = 0.361$) and no interaction ($p = 0.106$) between the mode of delivery and performance over the 5 trials (fig. 6f). However, due to the trend for a reduction in the latency to fall following a rest break (10 min between T2 and T3) in asphyxiated pups, data were analysed as the average latency to fall in pre-rest (T1 and T2) and post-rest (T3–T5) trials (fig. 6g). Two-way repeated-measures ANOVA (mode of delivery $\times$ pre-rest or post-rest trials) on these data showed a significant interaction ($p = 0.032$) and a significant effect of birth ($p = 0.036$). Subsequent post-hoc Bonferroni tests showed a significant difference ($p \leq 0.01$) between C-section and asphyxiated offspring in the post-break trials (fig. 6h).

Novel Object Recognition Test

The birth asphyxia offspring showed impaired learning and memory in this test at 28 days of age compared with C-section-delivered pups, in that they spent less time exploring a new object when placed back in the field 1 h after an initial learning trial of 30 min. The calculated discrimination index for the C-section pups was 0.55 ± 0.06 , compared to 0.38 ± 0.05 for the birth asphyxia pups ($p = 0.05$; fig. 7). There was no difference between the groups in the total time spent exploring the two objects during the recall trial, with asphyxiated pups spending 252 s and C-section pups spending 259 s exploring objects ($p = 0.8$), indicating that the new object was perceived as 'less novel', rather than with greater apprehension or anxiety. This is consistent with the observation reported above that, by day 28, pups from both groups spent similar times in the inner and outer zones of the open field.

Discussion

In this study, we have shown that birth asphyxia resulted in increased microglial activation and astrogliosis in the brain of the spiny mouse for at least 1 week after birth. A number of the brain regions affected in this way, including the hippocampus (memory and learning function), corpus callosum (integration of signals), inferior colliculus (auditory processing) and, to a lesser extent, areas of the cerebellum (motor coordination and balance), are consistent with an involvement in the cognitive deficits seen in children who were exposed to severe hypoxia in late gestation or to asphyxic conditions at birth [3–7]. Although the degree of actual asphyxia produced by retaining the fetuses in the ischemic uterus for the 7–8 min is unknown (we have found it impossible to obtain sufficient fetal blood for blood gas and pH analysis), the insult was relatively mild since most of the pups survived, and there was no evidence of cystic or focal lesions in the brain. Indeed, increased expression of the apoptotic protein caspase-3 was significant for only the corpus callosum and the molecular layer of the cerebellum.

Significant mortality followed in the cross-foster period due to maternal rejection or attack, or perhaps other factors which were not obvious as the animals were disturbed as little as possible to avoid interfering with mother-pup bonding. The failure rate was similar for both groups of pups, suggesting that the dams were not discriminating against post-asphyxia offspring in poor condition, as might have been expected. The relatively high rejection rate may, in part, be due to the prematurity of these pups, since they were delivered at day 37 (2 days prior to term), and we have noted that rejection by cross-foster dams is lower when pups are delivered at 38–39 days gestation [unpubl. obs.]. Postnatal growth of pups from both groups was similar, indicating that cross-foster care and nutrition for pups from the C-section and asphyxia groups was similar.

On postnatal days 1 and 7, the brains from asphyxiated offspring showed significant microglial activation and macrophage recruitment, which is a hallmark of brain injury previously described in human infants [for a review, see ref. 8]. The hippocampal regions (CA1 and CA3) and the inferior colliculus showed an increase in inflammatory cells which was apparent at day 1, and this persisted until at least day 7; the presence of activated microglia and macrophages at later ages was not examined in the current study. Activation of microglia and macrophage infiltration is associated with the necessary phagocytosis of cellular debris, but it also results in a burden of

increased production of neurotoxic substances, such as reactive nitrogen and oxygen species, pro-inflammatory cytokines, and the NMDA agonist quinolinic acid [for a review, see ref. 19]. Thus, prolonged activation of these monocyte-derived cells for at least a week may elicit further deleterious changes in the brain. That inflammatory reactions were progressing rather than resolving is indicated by the presence of microglial activation in some brain regions (corpus callosum, cerebellar molecular layer) on day 7 but not at day 1.

Astrogliosis has been noted in infants with perinatal asphyxia [10, 11], particularly in cases presenting with periventricular leucomalacia where astrocytes occupy the cavity left by focal necrotic lesions. In this study, where the birth asphyxia was not severe enough to induce focal cell death in white or grey matter, astrogliosis was still noted, suggesting that it might be a primary response of the developing brain to this type of insult. Astrogliosis was prominent in the corpus callosum at both 1 and 7 days of age, suggesting that astrocytes in this region either react immediately, or soon after, to the asphyxic event. In the CA1 region of the hippocampus, a prominent astroglial reaction was not evident until 7 days, again suggesting, as for the presence of activated microglia, that the response evolves in this brain region several days after the insult at birth.

The suggestion that reactions in the brain were progressing over several days after birth is also supported by the appearance of increased apoptosis in the corpus callosum and cerebellum. Apoptotic cell death is known to occur following asphyxia conditions as a result of impairment of mitochondrial function [9]. There was an increase in activated caspase-3 only in the corpus callosum at 1 day of age. Apoptosis occurs significantly later than necrotic cell death. Further studies should be conducted in order to assess the types of cells which were positive for caspase-3. It could be that the apoptotic cells at day 1 are not the same as those that express this apoptotic protein at day 7. A full histological examination of the postnatal development of the brain at 30 days and afterwards was beyond the scope of this study but is currently under investigation.

While it is apparent that the level of asphyxia experienced by these spiny mouse pups did not produce cystic or focal necrotic lesions in white matter, and that the damage in the brain appeared to be progressing rather than resolving until at least 1 week of age, it was sufficient to produce some motor impairment and deficits in complex behaviours involving learning, memory and, possibly, responses to novel situations that include stress and anxiety.

Locomotor activity can be affected by both physical and emotional factors. Reduced motivation, fatigue, physical impairment or difficulty with walking can reduce the total distance travelled. There was no significant difference in the total distance travelled by C-section-delivered or asphyxiated offspring in the open field at any postnatal age examined, suggesting the absence of gross motor deficits in the asphyxiated offspring. However, the need for several types of tests to assess locomotor abilities has been previously documented in mice due to the non-motor influences such as motivation, fatigue and higher cognitive functions [20]. Vestibulomotor tests assess fine-motor coordination – the ‘runged’ rotarod is an example of one such test for use in mice [21] – but this also has a cognitive component, as the animals may learn how to best perform the task. In the current study, the birth asphyxia offspring showed a significant reduction in the time they could remain on the rotarod (a finding that remained even after eliminating ‘best’ or ‘worst’ trials), but subsequent analysis revealed that impaired learning and memory may be involved since the improvement seen for pups from both groups over the first two trials was lost in the birth asphyxia pups when they were rested for 10 min between trials 2 and 3. In contrast, the C-section-delivered offspring tended to show continued improvement on the rotarod task following a rest period. It is unlikely that the inability to remain on the rotarod is a result of fatigue, as the latency to fall tended to increase with each subsequent trial at the start, or after the rest period. Thus, we conclude that impaired performance on the rotarod in the asphyxiated offspring is likely to be in at least in part attributed to impaired ability to learn and recall previously acquired information and apply it to a task. The results of the NORT also support the idea that learning and/or memory are affected in the asphyxia pups, as these offspring show a reduced discrimination index at 28 days of age. This particular test has been well characterized in rats and mice, and exploits the innate exploratory behaviour of rodents. The NORT was conducted in the open field in which the pups had been habituated every 3 days until day 28, and therefore the test of memory and learning was done in an environment with which the animal was familiar, thus minimizing the impact of stress and anxiety on the animal’s performance, which is a feature of the use of water mazes (and of particular significance for this species, as they have strong aversion to water). The reduction in the time spent exploring the novel object is unlikely to be attributable to reduced spontaneous locomotor activity as this was not evident at any age. A reduction in the *desire* to explore is unlikely to contribute

to the deficit, as there was no difference between C-section-delivered and asphyxiated offspring in the time spent exploring objects overall (novel plus familiar). Previous studies have demonstrated impaired spatial memory as a result of birth asphyxia [22], but our study suggests that birth asphyxia impaired both the ability to remember task-associated information (i.e., the rotarod), and non-spatial memory (the NORT). Our findings are consistent with a recent study of birth asphyxia in the term fetal rat (though, using 20 min of asphyxia), which showed impaired non-spatial memory using NORT, although the survival rate of the asphyxiated offspring was not reported and only data for male offspring were presented [22].

Although the interpretation of grooming behaviour in rodents is complex, changes in the incidence of this particular behaviour have been related to altered levels of anxiety [23], and thereby to perception of novelty and other stressors when animals are introduced into new environments and situations [24–26]. In the spiny mouse, we have frequently noted an increase of grooming when individual animals are removed from their ‘home’ cage for a treatment or for other reasons [unpubl. obs.]. It is important to note that in the current study, the spiny mice pups were placed in the open field every 3 days from day 1 to 28 of postnatal age, and thus habituation to the open field would be expected to occur during this period, so reducing anxiety-related behaviour. Indeed, while a significant difference in the amount of grooming was observed at 4 days of age, there was no difference in the number of bouts of grooming between C-section-delivered and asphyxiated offspring at 28 days. A number of studies using rats subjected to perinatal hypoxia also indicate that offspring have increased anxiety levels, although these tests were done at a significantly older age (1–2 years of age) [27, 28], and other studies show reduced anxiety at 2 months of age [29]. It may be necessary to

investigate anxiety more directly, using naïve animals in an open field, or protocols that provoke anxiety, such as the elevated-plus maze.

In summary, the current study demonstrates that impaired non-spatial and task-associated memory can arise as a result of a relatively short asphyxic insult at birth, in the absence of gross histopathological changes, and without severe motor impairments such as those that characterize cerebral palsy. Although the severity of the asphyxia, particularly in the brain itself, is unknown, it was sufficient to produce CNS inflammation and apoptotic cell death, which either persisted or continued to evolve for at least 1 week after the asphyxic event. The regions affected include those associated with memory and learning (hippocampus) and signal integration between hemispheres (corpus callosum). Further studies using this model of near-term birth asphyxia in a species in which the brain is more developed than in conventional rodents may allow the investigation of intervention strategies to determine whether the progression of brain injury can be arrested. Alternatively, the brain injury may be prevented by a maternal prophylactic treatment that protects the fetus against asphyxia-induced injury during the transition from fetus to neonate. The effect of such neuroprotective treatments can then be assessed not only on histopathological but, perhaps more importantly, on functional outcomes for the neonatal and juvenile offspring.

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